

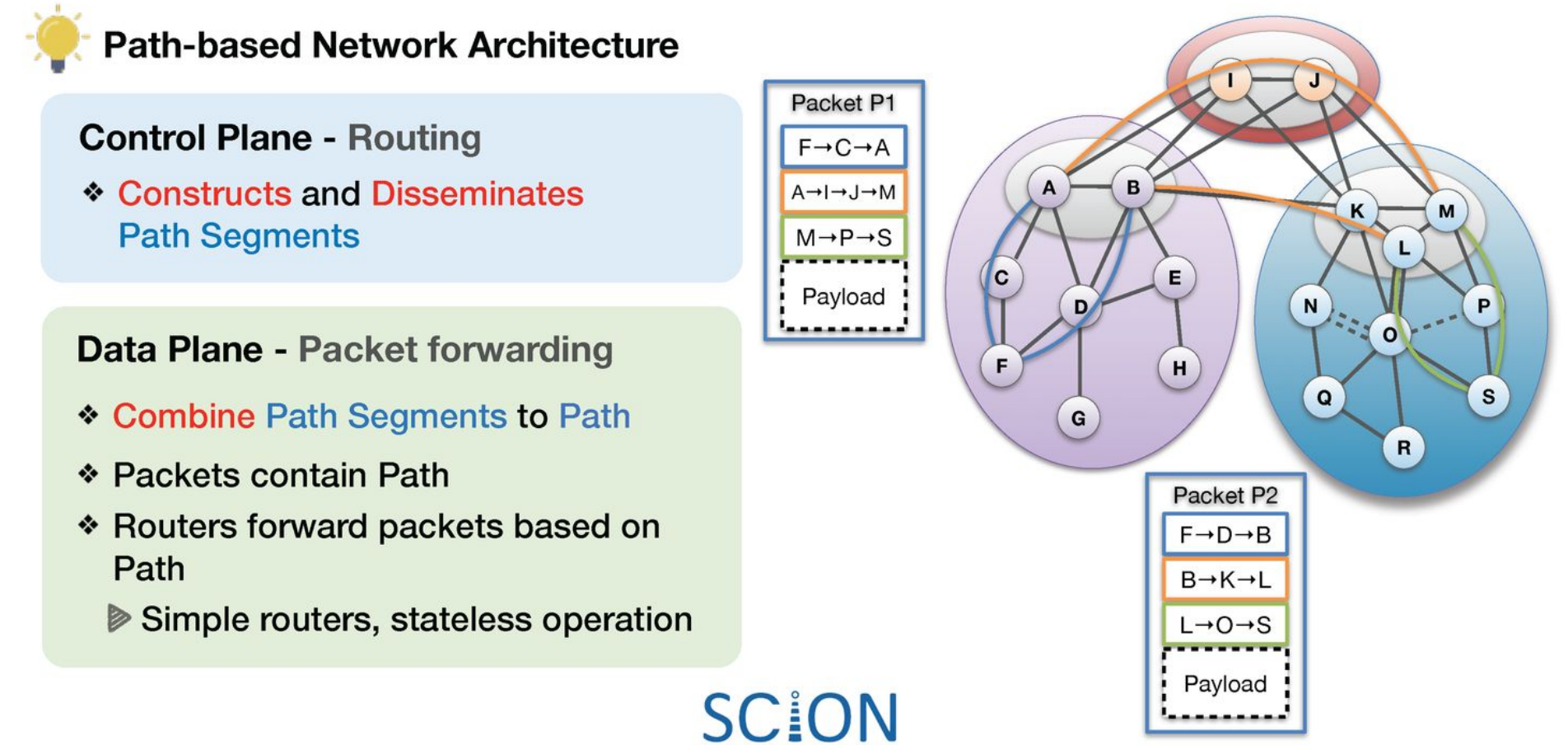
SCION Support for the SEED Emulator

SEED Emulator: A Python framework for emulating Internet topologies now includes native SCION support

Motivation

- Distributed nature of the Internet makes the evaluation of novel mechanisms and conducting realistic Internet-wide experiments a challenging task
- Especially testing the security of the Internet in an ethical manner without affecting productive systems is difficult.
- Network teaching often lacks practical examples, and virtualized testbeds for research are handcrafted with considerable effort.
- The SEcurity EDucation Emulator (SEED) provides an extensible, modular and well structured Python library to emulate Internet topologies and use-cases.
- To provide interested researchers a possibility to get easy and quick hands-on experience with SCION, we implemented SCION support for the SEED emulator, following the modular approach of the upstream codebase.

SCION Internet Architecture



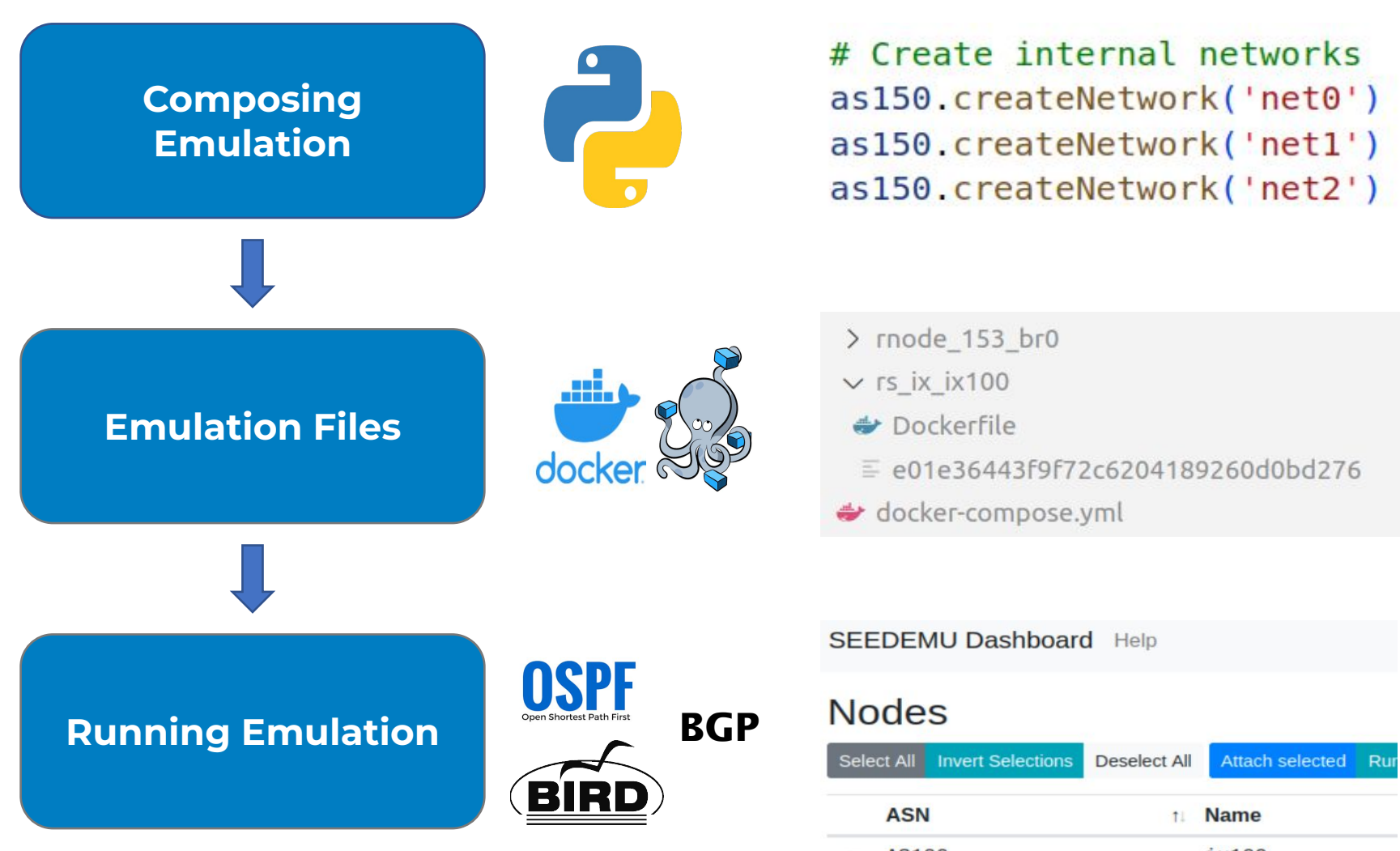
The SEED Emulator

- The goal is to refine educational methodologies and furnish learners with a direct, hands-on experience of Internet networking.
- Supports running the emulation on a single or multiple hosts, and also allows connecting to external hardware devices (e.g. switches, routers).
- Abstracts the codebase into extensible layers:
 - **Base:** Describes Autonomous Systems and hosts
 - **Routing:** Acts as the base of the routing protocols
 - **Ebgp:** Provides APIs for setting up External BGP peering
 - **Ibgp, Ospf:** Sets up full-mesh iBGP and OSPF routing within an AS
 - Layers for other protocols and applications (web servers, nameservers, etc.)

SCION Implementation in SEED

- Our implementation is designed to run SCION exclusively, or in parallel to eBGP. It follows SEED's modular and extensible structure.
- We provide the following layers to interact with the emulator:
 - **ScionBase:** Extends the Base layer by adding Isolation Domains
 - **ScionRouting:** Extends the Routing layer with SCION inter-AS routing
 - **ScionIsd:** Manages the trust relationships between Ases
 - **Scion:** Equivalent to the Ebgp layer for a BGP-based Internet; defines links between ASes
- We provide first SCION applications to run in SEED; more SCION applications are in progress to be added to SEED.

SEED Emulator Design



Example SCION Topology in SEED

```
# SCION ISDs
base.createIsolationDomain(1)

# Internet Exchange
base.createInternetExchange(100)

# AS-150
as150 = base.createAutonomousSystem(150)
scion_isd.addIsdAs(1, 150, is_core=True)
as150.createNetwork('net0')
as150.createControlService('cs1').joinNetwork('net0')
as150_router = as150.createRouter('br0')
as150_router.joinNetwork('net0').joinNetwork('ix100')
as150_router.crossConnect(153, 'br0', '10.50.0.2/29')
# Similar for 151, 152 and 153...

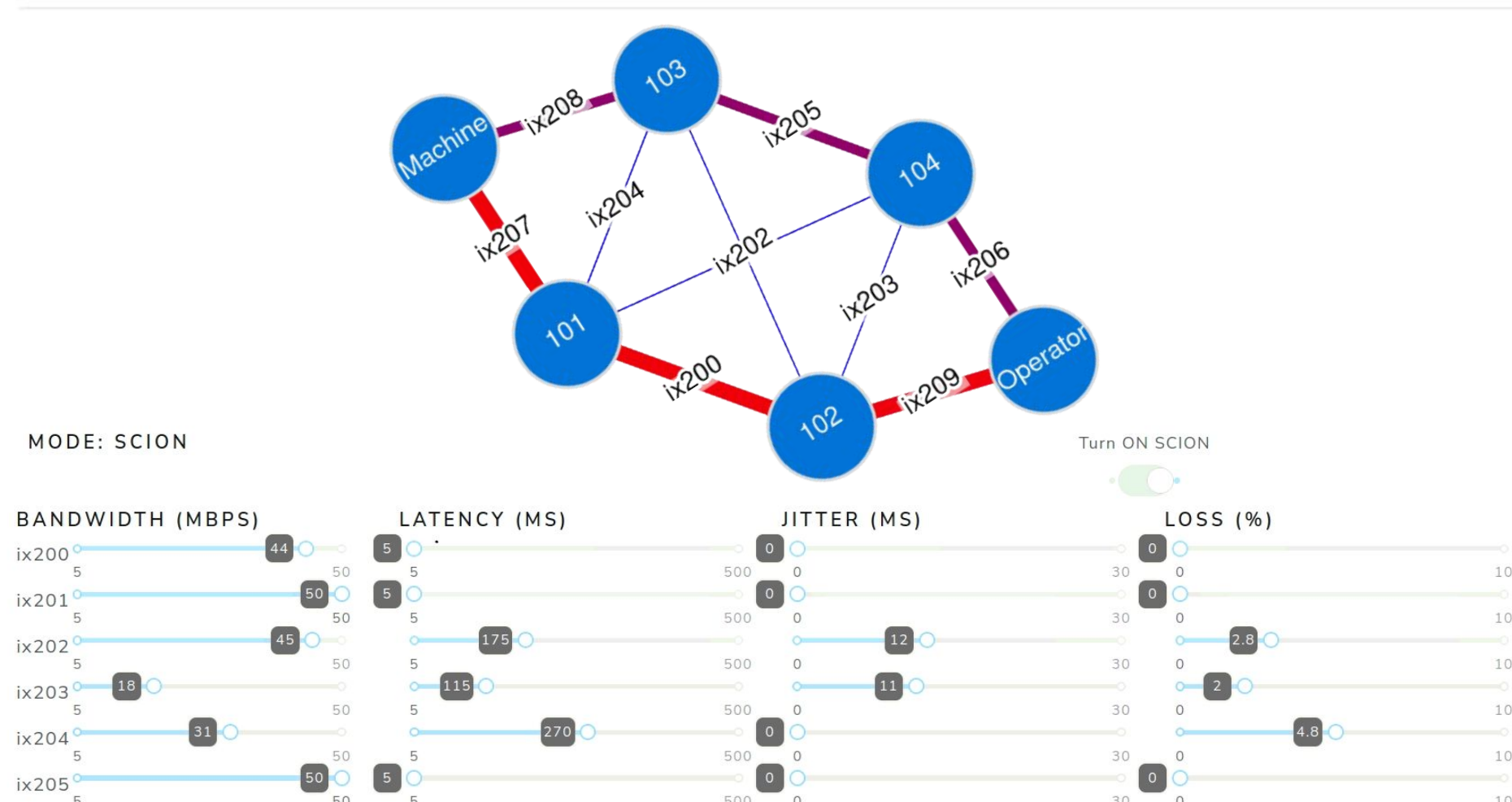
# Inter-AS routing
scion.addIxLink(100, (1, 150), (1, 151), ScLinkType.Core)
scion.addIxLink(100, (1, 151), (1, 152), ScLinkType.Core)
scion.addIxLink(100, (1, 152), (1, 150), ScLinkType.Core)
scion.addXcLink((1, 150), (1, 153), ScLinkType.Transit)
```



Check the Code!

SCION over SEED in Action

NETWORK VISUALIZATION



Contact us

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Research and Education Network

- We are running the SCION Education, Research and Academic (SCIARA) network to connect universities and national research and education networks (NRENs).
- We provide native SCION connectivity to students and researchers, in an overlay-free, productive network spanning over three continents.
- The following Figure depicts the deployment status of the network, core-ASes are marked in blue, dashed lines indicate that the nodes and links are currently being set up.

